

Car Price Prediction

Using Regularized Regression

Applying Ridge (L2), Lasso (L1), and ElasticNet to Automotive Pricing Data

Author	MBulah
Module	Machine Learning — Regression
Dataset	CarPrice_Assignment.csv
Records	205 rows × 26 columns
Target Variable	price (USD)
Submission Date	June 2026

Project Objective

This report develops and benchmarks four regression models — Ordinary Least Squares (OLS), Ridge, Lasso, and ElasticNet — to estimate automobile selling prices in the US market from vehicle specifications and manufacturer information. The analysis addresses a practical market-entry challenge: identifying which measurable attributes most strongly govern pricing, and quantifying prediction accuracy under different regularization strategies. The broader aim is to equip decision-makers with data-driven insight into the key pricing levers within the US automotive sector.

Table of Contents

Section	Page
1. Introduction & Problem Statement	3
2. Dataset Overview	3
3. Exploratory Data Analysis (EDA)	4
3.1 Price Distribution	4
3.2 Feature Engineering — Car Manufacturer	4
3.3 Categorical Attributes vs. Price	5
3.4 Correlation Analysis	5
3.5 Scatter Plots — Top Features vs. Price	6
3.6 Outlier Detection	6
4. Data Preprocessing	7
4.1 Encoding Categorical Variables	7
4.2 Train-Test Split & Feature Scaling	7
5. Multicollinearity — VIF Analysis	8
6. Model Training	8
6.1 Ordinary Linear Regression (Baseline)	8
6.2 Ridge Regression (L2 Penalty)	9
6.3 Lasso Regression (L1 Penalty)	9
6.4 ElasticNet Regression	10
7. Model Evaluation & Comparison	10
8. Insights, Assumptions & Recommendations	12
9. References	13

1. Introduction & Problem Statement

The US automotive market is among the most dynamic and competitive in the world, shaped by a complex interplay of engineering specifications, brand positioning, fuel efficiency regulations, and consumer preferences. For a manufacturer considering entry into this market, understanding how observable vehicle attributes translate into price expectations is a strategic necessity.

This study approaches that challenge quantitatively. Using a curated dataset of 205 US-market vehicles, four supervised regression models are trained, tuned, and compared: a standard OLS baseline alongside three regularized variants — Ridge (L2), Lasso (L1), and ElasticNet. Regularization is of particular interest here because the feature matrix — after encoding categorical variables — expands to 64 dimensions, creating conditions where ordinary least-squares can overfit and multicollinearity can distort coefficient estimates. Ridge dampens all coefficients proportionally; Lasso drives many to exactly zero, performing embedded feature selection; ElasticNet blends both behaviours. Comparing these methods reveals not only which generalizes best, but also which vehicle attributes most consistently drive price.

2. Dataset Overview

The **CarPrice_Assignment.csv** dataset comprises 205 records and 26 fields describing vehicle characteristics and the corresponding retail price in USD. No missing values were detected, eliminating the need for imputation. The target variable is **price**; all remaining columns serve as candidate predictors.

Table 1: Feature Inventory

Feature	Type	Description
car_ID	Integer	Unique record identifier (dropped before modelling)
symboling	Integer	Actuarial risk rating from -3 (safe) to +3 (risky)
CarName	String	Full vehicle name — make and model combined
fueltype	Categorical	Fuel variant: gas or diesel
aspiration	Categorical	Induction method: standard or turbo
doornumber	Categorical	Door count: two or four
carbody	Categorical	Body style — sedan, hatchback, convertible, etc.
drivewheel	Categorical	Driven axle: fwd, rwd, or 4wd
enginelocation	Categorical	Engine placement: front or rear
wheelbase / carlength / carwidth / carheight	Float	Physical dimensions (inches)
curbweight	Integer	Unloaded vehicle mass (lbs)
enginetype	Categorical	Engine architecture — ohc, dohc, rotor, etc.
cylindernumber	Categorical	Number of cylinders
enginesize	Integer	Displacement in cubic inches
fuelsystem	Categorical	Fuel delivery mechanism

Feature	Type	Description
boreratio / stroke	Float	Bore-to-stroke geometry ratios
compressionratio	Float	Engine compression ratio
horsepower / peakrpm	Integer	Power output and redline speed
citympg / highwaympg	Integer	Fuel economy in city and highway conditions
price	Float	TARGET — retail price in USD

Table 2: Descriptive Statistics — Key Numerical Features

Statistic	price	enginesize	horsepower	curbweight	citympg	highwaympg
Count	205	205	205	205	205	205
Mean	\$13,277	126.9	104.3	2,556	25.2	30.8
Std Dev	\$7,989	41.6	39.5	521	6.5	6.9
Min	\$5,118	61	48	1,488	13	16
Median	\$10,295	120	95	2,414	24	30
Max	\$45,400	326	288	4,066	49	54

3. Exploratory Data Analysis (EDA)

Prior to modelling, the dataset was examined thoroughly to characterise variable distributions, uncover inter-feature relationships, and flag potential data quality concerns. The EDA findings inform all subsequent preprocessing and modelling choices.

3.1 Price Distribution

Retail prices span from \$5,118 to \$45,400, with a mean of \$13,277 and a median of \$10,295 — a gap that reflects positive skewness (skewness ≈ 1.63). The bulk of vehicles cluster at the lower end of the range, with a long right tail driven by a small number of high-end and exotic models. Applying a log transformation $\log(\text{price} + 1)$ brings the distribution closer to symmetry, which may benefit models that assume normally distributed residuals. For this study, untransformed prices were used throughout to keep results directly interpretable in USD.

3.2 Feature Engineering — Extracting Car Manufacturer

The raw **CarName** column encodes both brand and model in a single free-text string. The manufacturer label was isolated by splitting on whitespace and retaining the leading token, followed by lowercasing for consistency. A manual audit revealed five typographic variants that mapped to existing brands and were corrected accordingly:

- maxda → mazda
- toyouta → toyota
- vokswagen and vw → volkswagen
- porcshce → porsche
- alfa-romero → alfa-romeo

After cleaning, 22 distinct manufacturers are represented. Toyota accounts for the most entries (32), followed by Nissan (18) and Mazda (17). Despite low record counts, Jaguar, Porsche, and BMW achieve the highest average transaction prices, confirming that brand identity carries substantial pricing power independent of specification-level features.

3.3 Categorical Attributes vs. Price

Average selling prices were computed for each level of the six primary categorical features. The following patterns emerged:

- **Fuel type:** Diesel-powered vehicles average higher prices than their petrol counterparts, likely reflecting the diesel premium in this era of the dataset.
- **Aspiration:** Turbocharged models carry a meaningful price premium over naturally aspirated ones, consistent with their higher performance output.
- **Door count:** Two-door configurations — typically sportier coupés — average higher prices than four-door variants.
- **Body style:** Hardtops and convertibles occupy the top of the price hierarchy; hatchbacks and wagons sit at the lower end.
- **Drive wheel:** Rear-wheel-drive vehicles command substantially higher average prices than front-wheel-drive, reflecting their typical association with performance and luxury segments.
- **Engine location:** Rear-engine placement — confined to a handful of high-performance models — correlates with the highest average prices in the dataset.

3.4 Correlation Analysis

Pearson correlations were computed between all numerical predictors and the target variable. The findings below identify which continuous attributes carry the strongest linear signal for price prediction.

Table 3: Pearson Correlation of Numerical Features with Price

Feature	Correlation with Price	Strength & Direction
enginesize	0.87	Strong positive
curbweight	0.84	Strong positive
horsepower	0.81	Strong positive
carwidth	0.76	Strong positive
carlength	0.68	Moderate positive
wheelbase	0.58	Moderate positive
highwaympg	-0.70	Strong negative
citympg	-0.69	Strong negative
symboling	-0.08	Negligible
stroke	0.08	Negligible

Engine displacement, kerb weight, and power output emerge as the three strongest positive correlates. Fuel-economy metrics show strong negative associations — larger, more powerful vehicles consume more fuel and command higher prices, while economy-oriented models tend to be lighter and cheaper. This inverse relationship between efficiency and cost is a consistent structural feature of this dataset.

3.5 Scatter Plots — Top Features vs. Price

Bivariate scatter plots for the six highest-correlated features confirm broadly linear relationships with price, though some heteroscedasticity (widening spread at higher price points) is visible. This is consistent with the right-skewed distribution of the target and suggests that prediction errors may be proportionally larger for premium vehicles.

3.6 Outlier Detection

Box plots were produced for each of the six key numerical features. Enginesize and horsepower exhibit the most pronounced upper-tail outliers, corresponding to high-displacement, high-output engines found in sports and luxury models. These observations are genuine rather than erroneous and were retained in all subsequent analyses, as removing them would artificially suppress the range of vehicle types the models must handle.

4. Data Preprocessing

The raw encoded dataset was transformed into a clean, model-ready feature matrix through three sequential steps: categorical encoding, data partitioning, and feature standardisation.

4.1 Encoding Categorical Variables

Two encoding strategies were applied based on feature cardinality:

- **Binary (label) encoding** was used for the four features with exactly two categories: fueltype (gas=1, diesel=0), aspiration (turbo=1, std=0), doornumber (two=1, four=0), and enginelocation (front=1, rear=0). This approach is appropriate because neither ordering nor distance assumptions are violated for binary attributes.
- **One-hot encoding** with *drop_first=True* was applied to all remaining multi-class categoricals: carbody (5 classes → 4 dummies), drivewheel (3→2), enginetype (7→6), cylindernumber (7→6), fuelsystem (8→7), and carcompany (22→21). Dropping one dummy per group eliminates perfect collinearity with the intercept term.

The identifiers `car_ID` and `CarName` were removed entirely (the latter replaced by the engineered `carcompany` field). The resulting feature matrix contains **64 columns**.

Table 4: Encoding Strategy Summary

Method	Features	Rationale
Binary / Label	fueltype, aspiration, doornumber, enginelocation	Exactly two levels; label encoding is ordinal-safe
One-Hot (drop first)	carbody, drivewheel, enginetype, cylindernumber, fuelsystem, carcompany	Multiple levels; prevents perfect collinearity with intercept
Dropped	car_ID, CarName	Identifier / superseded by carcompany

4.2 Train-Test Split and Feature Standardisation

The 205 samples were divided into an 80% training set (164 observations) and a 20% held-out test set (41 observations) using `random_state=42` for full reproducibility. **StandardScaler** was fitted exclusively on the training data and then applied to transform both partitions — a critical discipline that prevents any test-set statistics from leaking into model training. Standardisation is mandatory for Ridge, Lasso, and ElasticNet: because the penalty term sums over all squared or absolute coefficients, features on larger raw scales would otherwise attract disproportionately large penalties, distorting the regularization effect.

Table 5: Partitioning and Scaling Configuration

Parameter	Value
Total observations	205
Training set	164 samples (80%)
Test set	41 samples (20%)
Random state	42
Scaler applied	StandardScaler — zero mean, unit variance
Feature columns (post-encoding)	64

5. Multicollinearity — Variance Inflation Factor (VIF)

Variance Inflation Factor quantifies the degree to which each predictor's variance is inflated as a consequence of its linear correlation with the remaining features. A VIF score above 10 signals a level of collinearity that can render OLS coefficient estimates unstable and difficult to interpret. VIF was computed on the base numerical feature set prior to one-hot encoding.

Table 6: VIF Scores for Key Numerical Predictors

Feature	VIF Score	Classification
fueltype	75.15	Severe — critically collinear
compressionratio	63.40	Severe — critically collinear
curbweight	13.34	High — collinear (VIF > 10)
carlength	9.41	Moderate — borderline
wheelbase	8.81	Moderate
horsepower	7.70	Acceptable
enginesize	6.71	Acceptable
carwidth	5.21	Acceptable
aspiration	2.63	Low
doornumber	2.58	Low

The severe VIF scores for fueltype and compressionratio reflect their strong linear dependency on combinations of other variables. While these features are retained for completeness, the presence of multicollinearity strongly motivates the use of regularized estimators: Ridge and Lasso constrain coefficient magnitudes, redistributing predictive weight more stably across correlated predictors rather than concentrating it erratically on any one feature.

6. Model Training

Four regression models were fitted to the standardised training data. Each model is described below with its objective function and the key hyperparameter choices made.

6.1 Ordinary Linear Regression — Baseline

Standard OLS minimises the residual sum of squares with no constraint on coefficient size. Its closed-form solution is computationally efficient but can overfit when the predictor count approaches the sample count (here 64 features, 164 training samples). Its test-set performance establishes the benchmark against which all regularized variants are judged.

Objective: Minimise $L = \sum (y_i - \hat{y}_i)^2$

6.2 Ridge Regression — L2 Penalty

Ridge regression appends a squared-coefficient penalty to the OLS objective, controlled by the hyperparameter α . This shrinks all coefficients toward — but never exactly to — zero, distributing predictive weight across correlated features and reducing variance at the cost of a small increase in bias. The regularization strength was chosen via 5-fold cross-validation across a logarithmic grid {0.01, 0.1, 1, 10, 50, 100, 200, 500, 1000}; $\alpha = 10$ yielded the highest mean CV R^2 .

Objective: Minimise $L = \sum (y_i - \hat{y}_i)^2 + \alpha \sum \beta_j^2$

6.3 Lasso Regression — L1 Penalty

Lasso replaces the squared penalty with an absolute-value term. This geometric change has a crucial practical consequence: the L1 constraint introduces corners in the feasible set, and the optimal solution frequently lands at a corner where one or more coefficients equal exactly zero. Lasso therefore simultaneously estimates and selects features, producing a sparse model that is easier to interpret. With 64 input features — many of which are one-hot dummies for rarely occurring categories — this sparsity is operationally valuable. $\alpha = 50$ was adopted following exploratory cross-validation.

Objective: Minimise $L = \sum (y_i - \hat{y}_i)^2 + \alpha \sum |\beta_j|$

6.4 ElasticNet Regression — Combined L1 + L2 Penalty

ElasticNet combines both penalties in a single objective governed by two parameters: α controls overall penalty strength and $l1_ratio$ governs the L1-to-L2 mixture (0 = pure Ridge; 1 = pure Lasso). With $l1_ratio = 0.5$ and $\alpha = 50$, equal weight is given to each penalty, targeting a balance between feature selection and coefficient shrinkage. ElasticNet is particularly advantageous when groups of correlated features should be selected or suppressed together — a characteristic of this dataset's manufacturer dummies.

Objective: Minimise $L = \sum (y_i - \hat{y}_i)^2 + \alpha [l1_ratio \sum |\beta_j| + (1-l1_ratio) \sum \beta_j^2]$

7. Model Evaluation & Comparison

All four models were assessed on the 41-observation test set — data withheld entirely from training and hyperparameter selection. Three complementary metrics were computed:

- **RMSE (Root Mean Squared Error):** The square root of average squared deviations between actual and predicted prices, expressed in USD. Larger errors are penalised disproportionately, making RMSE sensitive to outlier predictions.
- **MAE (Mean Absolute Error):** The arithmetic mean of absolute deviations. More robust than RMSE to individual large errors; also reported in USD for direct interpretability.
- **R² (Coefficient of Determination):** The proportion of price variance explained by the model. A value of 1.0 indicates perfect fit; 0.0 indicates the model performs no better than predicting the training mean for every observation.

Table 7: Test-Set Performance Summary

Model	RMSE (USD)	MAE (USD)	R ²	Rank
Linear Regression (OLS)	\$2,669.93	\$1,763.57	0.9097	1st
Lasso ($\alpha=50$)	\$3,040.84	\$2,087.44	0.8829	2nd
Ridge ($\alpha=10$)	\$3,343.19	\$2,194.06	0.8584	3rd
ElasticNet ($\alpha=50$)	\$7,777.67	\$5,474.40	0.2337	4th

Table 8: Regression Metric Definitions

Metric	Formula	Interpretation
RMSE	$\sqrt{\text{mean}((y - \hat{y})^2)}$	Error in same units as target (USD); sensitive to large prediction errors
MAE	$\text{mean}(y - \hat{y})$	Average absolute error in USD; more robust to outlier predictions than RMSE
R ²	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$	Fraction of price variance explained; 1.0 = perfect, 0.0 = naive mean prediction

Interpretation of Results

OLS performed best overall ($R^2 = 0.910$, $\text{RMSE} = \$2,670$), explaining approximately 91% of variation in test-set prices. This result is notable given the acknowledged multicollinearity and dimensionality of the feature space; the 80/20 split apparently provides the OLS estimator sufficient training data to generalise without overfitting severely on this particular test partition.

Lasso ranked second ($R^2 = 0.883$, $\text{RMSE} = \$3,041$). By zeroing out the coefficients of many one-hot dummies for infrequent manufacturers and minor body-style variants, it produced a more parsimonious model that remained highly predictive. From an interpretability standpoint, Lasso's sparse solution is arguably more actionable than OLS, because it clearly delineates which attributes drive pricing and which do not.

Ridge ranked third ($R^2 = 0.858$, $\text{RMSE} = \$3,343$). Unlike Lasso, Ridge retains all features but reduces their collective influence, which can be sub-optimal when many predictors are genuinely irrelevant — a situation likely present here given the breadth of one-hot encoded categories.

ElasticNet underperformed substantially ($R^2 = 0.234$, $\text{RMSE} = \$7,778$) with the chosen $\alpha = 50$. The combined penalty at this strength appears excessively aggressive, shrinking too many coefficients toward zero. A systematic grid search over both α and l1_ratio would be required to evaluate ElasticNet's true

potential on this dataset.

Residual plots for OLS and Lasso show errors broadly centred on zero across the prediction range, with some widening at higher predicted prices — consistent with the positively skewed target distribution. This heteroscedasticity suggests that modelling log-transformed prices could improve calibration at the upper end of the market.

8. Insights, Assumptions & Recommendations

8.1 Key Analytical Insights

- **Strongest pricing signals:** Engine displacement, kerb weight, power output, and vehicle width are the four attributes most positively correlated with price. City and highway fuel economy carry the strongest negative associations — a structural reflection of the inverse relationship between vehicle size and efficiency in this dataset.
- **Brand premium:** Manufacturer identity exerts a pricing influence that goes beyond what specification-level features alone can explain. Luxury and performance brands (Jaguar, Buick, Porsche, BMW) consistently command premiums that persist even after controlling for engine and dimension variables.
- **Multicollinearity is significant:** VIF values above 60 for fueltype and compressionratio, and above 13 for curbweight, confirm that OLS coefficient estimates for individual features should be interpreted cautiously. Regularized alternatives (Ridge, Lasso) handle this more robustly.
- **Model selection trade-off:** OLS achieves the highest raw accuracy on the test set, but Lasso offers only marginally lower accuracy while delivering a sparse, interpretable solution. For a business use case where identifying *which* attributes matter is as important as prediction accuracy, Lasso is arguably the superior choice.
- **ElasticNet requires tuning:** The default hyperparameter settings applied here are too restrictive. A cross-validated search over a grid of α and l1_ratio values is necessary before drawing conclusions about ElasticNet's viability for this problem.

8.2 Modelling Assumptions

- No missing values were present in the raw dataset; imputation was unnecessary.
- The car manufacturer was assumed to be the first whitespace-delimited token in the CarName field, confirmed through manual inspection of the resulting brand list.
- Binary encoding is ordinal-safe for two-category features like doornumber, as no meaningful ordinal relationship exists between the two classes.
- One-hot encoding with drop_first=True prevents perfect multicollinearity between dummy columns and the model intercept.
- StandardScaler was fitted on the training partition only and applied to both train and test sets, ensuring no data leakage from test statistics into model training.
- random_state=42 and an 80/20 split ratio were used consistently across all experiments to ensure comparable conditions.
- For ElasticNet, l1_ratio=0.5 was used as an initial starting point; this parameter was not cross-validated.
- High-valued outliers in enginesize and horsepower were treated as valid observations representing genuine high-performance vehicles rather than recording errors, and were retained in all analyses.
- Ridge alpha tuning used 5-fold cross-validation applied exclusively to the training set; no test-set information influenced this choice.
- All prices are denominated in USD; no currency conversions were applied.

8.3 Recommendations for Further Development

- Apply a log transformation to the target variable to address positive skewness and potentially reduce heteroscedasticity in residuals, particularly at high price points.

- Consider removing or combining fueltype and compressionratio with other features prior to OLS fitting, given their severe VIF scores. This may stabilise coefficient estimates without sacrificing predictive accuracy.
- Conduct a full grid search over both α and l1_ratio for ElasticNet to assess its true performance ceiling relative to the single-penalty models.
- Benchmark the linear models against tree-based alternatives — Random Forest and Gradient Boosting — which handle non-linearity and feature interactions naturally and may outperform on high-dimensional mixed-type datasets.
- For actionable business guidance, the Lasso coefficient profile (identifying which specific manufacturer dummies, body styles, and engineering specs survive regularization) should be communicated to product planning teams as a data-driven pricing framework.

9. References

The following sources were consulted for conceptual background on the methods applied in this study. All analytical decisions, written interpretations, and conclusions presented in this report are original.

[1] Scikit-learn Development Team. (2024). Linear Models — scikit-learn documentation.

https://scikit-learn.org/stable/modules/linear_model.html

[2] GeeksForGeeks. (2024). Ridge Regression vs Lasso Regression in Machine Learning.

<https://www.geeksforgeeks.org/ridge-regression-vs-lasso-regression/>

[3] DataCamp. (2024). Lasso and Ridge Regression in Python Tutorial.

<https://www.datacamp.com/tutorial/tutorial-lasso-ridge-regression>

[4] GeeksForGeeks. (2024). One-Hot Encoding vs Label Encoding — explained.

<https://www.geeksforgeeks.org/one-hot-encoding-vs-label-encoding/>

[5] GeeksForGeeks. (2024). Feature Scaling: Normalisation and Standardisation.

<https://www.geeksforgeeks.org/feature-engineering-scaling-normalization-and-standardization/>

[6] GeeksForGeeks. (2024). Regression Metrics for Machine Learning.

<https://www.geeksforgeeks.org/regression-metrics/>

[7] Kumawat, S. (2023). Performance Metrics for Regression. Medium.

https://medium.com/@Sunil_Kumawat/performance-metrics-for-regression-39d43d9ba2e0

[8] GeeksForGeeks. (2024). Detecting Multicollinearity with VIF in Python.

<https://www.geeksforgeeks.org/detecting-multicollinearity-with-vif-python/>

[9] SampattKumar. (2023). Linear Regression — Car Dataset. GitHub.

<https://github.com/SampattKumar/Linear-Regression-Car-Dataset>